



Gun Cathode Conditioning System User's Manual

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Canadian Light Source Inc.
101 Perimeter Road
University of Saskatchewan
Saskatoon, Saskatchewan
S7N 0X4 Canada

Signature

Date

Original on File – Signed by:

Author

Karen Vanthuyne

Approver

Neil Johnson

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Revision History

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1.0 PURPOSE

This document outlines the gun cathode conditioning process which is installed in R1023-06. The system is used to condition gun cathodes in advance so that when a replacement is needed in the linac, little time is sacrificed. The process is mainly automated and, depending on various factors, takes about one week to complete.

2.0 OVERVIEW OF THE SYSTEM

The process begins by pumping down the vacuum chamber. This involves using a turbo pump for the rough pumping and once the roughing port valve is closed, turning on the ion pump. When the vacuum level is adequate the ON push button is pressed and the system will begin stepping up the filament voltage. The system will wait for one hour at Good Vacuum and then the filament voltage will increase by 0.1V. Another hour of Good Vacuum is required after each 0.1V increase. If the system rises above Intermediate Vacuum, the voltage will be stepped down in the same steps. The increasing process will continue once the vacuum re-enters the Good Vacuum state. Once the filament voltage reaches 12.5V the anode voltage will begin a similar process, ramping up or down in steps of 0.1kV. The maximum anode voltage is 2kV, so once this value is met the system is at maximum voltage. The goal of the system is to see a current of at least 0.82mA between the anode and the cathode. ^[1] This current is continually monitored throughout the process. Things such as bad vacuum and the OFF push button will trigger a shutdown of the whole system.

3.0 EQUIPMENT NUMBERS

- PS1023-01 – Anode Power Supply – Glassman EH2P50L
- PS1023-02 – Filament Power Supply – Kepco BOP 20-5D
- PS1023-03 – Ion Pump Controller – Varian Dual Controller
- PS1023-04 – PLC 24VDC Power Supply
- TCG1023-01 – Thermocouple Gauge - Varian
- CCG1023-01 – Cold Cathode Gauge - Varian
- VMGC1023-01 – Varian Multi-gauge Controller
- TCGC1023-01 – Thermocouple Gauge Controller - Varian
- CCGC1023-01 – Cold Cathode Gauge Controller - Varian
- IOP1023-01 – Ion Pump – Varian 150, Diode
- PLC1023-02 – 171CCC96030 processor adapter with 2 170AMM09000 modules
- R1023-06 – location of the system
- DPNL1023-01-CCT2 – Source

4.0 PLC INPUTS AND OUTPUTS

This system uses a 171CCC96030 processor adapter along with two 170AMM09000 modules. There are four analog inputs, two analog outputs, four discrete inputs, and two discrete outputs on each module.

On Base #1, the analog inputs are used to monitor the filament and anode voltages and currents. The analog outputs are used to send signals to the filament and anode power supplies. On Base #2, three analog inputs are used. One receives from the ion pump controller, and the other two from the multi-gauge controller reading CCG and TCG information.

The discrete inputs of Base #1 are used by contacts on the CCG and TCG as well as the ON and OFF push buttons located on the top panel of the rack. Also on this top panel are four LEDs labeled 'System "ON"', 'Vacuum Fault', 'Filament Run-up Complete', and 'Conditioning Complete'. The first two LEDs are controlled by the discrete outputs in Base #1, the second two by Base #2.

5.0 VACUUM STATES

There are three vacuum set points;

-1×10^{-7} Torr	Bad Vacuum
-5×10^{-8} Torr	Intermediate Vacuum
-1×10^{-8} Torr	Good Vacuum

The vacuum spectrum can be divided into four states;

Less than or equal to Good Vacuum

When the vacuum is less than or equal to 1×10^{-8} it is considered a good vacuum. Therefore it is safe to increase the filament or anode power supplies by 0.1V or 0.1kV respectively, waiting for one hour between increments to ensure Good Vacuum.

Greater than Good Vacuum but less than or equal to Intermediate Vacuum

With the vacuum in this stage the power supplies will neither increase nor decrease their voltages. At this point the system waits and monitors the vacuum. Appropriate action will be taken depending on whether the vacuum increases or decreases, entering a new vacuum state.

Between Bad Vacuum and Intermediate Vacuum Exclusively

If the vacuum rises above the Intermediate Vacuum set point, action needs to be taken to restore good vacuum. The voltage is immediately stepped down by 0.1V or 0.1kV intervals each minute until the vacuum lowers again.

Greater than or equal to Bad Vacuum

When the vacuum reads greater than or equal to Bad Vacuum, the vacuum is considered dangerously high and the whole system is shut down. Both the filament and the anode power supplies are immediately reduced to zero and the system must be manually restarted.

6.0 VOLTAGE RAMPING

The ON push button is pressed.

If the system is in the Good Vacuum state the ramping process will begin. If the system is in the Bad Vacuum state the ramping process will not begin and the 'Vacuum Fault' LED will be lit. Once the 'Vacuum Fault' LED is off the vacuum status is below Bad Vacuum and the ON button is pressed again.

The 'System "ON"' LED lights and the filament voltage begins ramping up. An hour of Good Vacuum must be held before the filament voltage increases. It is increased in steps of 0.1V and after each successful step up, an hour of Good Vacuum is required before stepping up again.

If the vacuum rises above the Good Vacuum set point yet not above the Intermediate Vacuum set point the system will neither increase nor decrease the voltage and will wait until the vacuum status moves into a different vacuum state. If the vacuum improves to Good Vacuum, the filament voltage will continue the increasing process. If the vacuum rises to above Intermediate Vacuum the voltage will begin to decrease.

As soon as the vacuum rises above the Intermediate Vacuum set point, the filament voltage decreases by 0.1V. The minute timer is set, and until the vacuum lowers to at least the Intermediate Vacuum set point, the filament voltage will be stepped down by 0.1V per minute. If the vacuum status shows Good Vacuum the hour timer will run and if the vacuum stays good for the full length of time, the filament voltage will begin the increasing process.

Once the filament voltage has reached 12.5V, the 'Filament Run-up Complete' LED will light and the anode voltage will begin stepping up. The anode voltage is ramped up or down in intervals of 0.1kV. It follows the same vacuum set points and timing specifications as the filament voltage ramping. The anode voltage will not exceed 2kV. If the anode voltage is reduced to 0V during the ramping down process and the vacuum is still between Bad Vacuum and Intermediate Vacuum, the system will resort to ramping down the filament voltage from 12.5V until the vacuum status enters a different state. If the vacuum returns to Good Vacuum the ramping up process will begin where the ramping down ended.

If at any point in this process the vacuum status enters the Bad Vacuum region, the whole system will be shut down. Both power supplies will be set to 0V and the system will need to be restarted manually. The 'Vacuum Fault' LED will light, signifying a vacuum error and the 'System "On"' LED will turn off, signifying the system has been shut down. The Bad Vacuum trip can be set off by the analog input readings of the TCG and CCG, the ion pump controller input reading, or the TCG and CCG discrete contacts. The system can also be shut down by pressing the OFF push button.

Throughout the ramping up process, the anode current is monitored. The anode voltage will not increase past 2kV, so once it reaches this point the filament and anode voltages are held constant, pending good vacuum, and we wait for the anode current to reach at least 0.82mA. If the anode current should reach this value before the peak of the power supplies, the ramping

process will stop and the voltages will remain where they are. A 24 hour timer begins when at least 0.82mA of current flows between the anode and the cathode. The current will run for this length of time, while the filament and anode voltages remain constant. When the timer is up both power supplies will be immediately decreased to output 0V and the cathode will cool down for another 24 hours. Once this last timer is up, the 'Conditioning Complete' LED will light to signify that the entire process is complete.

7.0 REFERENCES

- 1) Haimson Research Corporation. University of Saskatchewan, 13 MeV Injection Linear Accelerator MODEL HRC-735 Instruction and Maintenance Manual. August 1980. pg 127.
- 2) P1023006
- 3) F:\TST\PID\0145520
- 4) F:\TST\EE\WIR\0145521-23
- 5) F:\TST\EE\PLC\WIR\0145524-26
- 6) F:\TST\EE\WIR\0145527
- 7) F:\CDAC\EE\0145528